A REPORT ON

ON JOB TRAINING (OJT)

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE)

Semester V

By

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Under the Guidance of

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Department of Computer Science

DEVCHAND COLLEGE

Arjunnagar

Affiliated to

SHIVAJI UNIVERSITY, KOLHAPUR

Academic Year: [2026–2027]

CERTIFICATE FROM THE ORGANIZATION / COMPANY

[Organization Letterhead / Logo]

This is to certify that **Miss. Snehal Yuvraj Magdum**, a student of B.Sc. III (Computer Science), Devchand College Arjunnagar, has successfully completed On Job Training at YBI Foundation during the period from 23/07/2026 to 23/09/2026, for a total duration of 8Weeks.

During the training, the trainee worked on [PROJECT / DOMAIN / TASKS, e.g., "web application development, deepfake/AI model assistance, software testing"] and demonstrated [sincerity, discipline, technical aptitude, etc.]. Her conduct and performance during the training period was found to be satisfactory / good / excellent.

We wish him/her success in all future endeavours.

Authorized Signatory

Name & Designation: [_____]

Organization Name & Seal: [_____]

Date: [_____] Place: [_____]

Devchand College, Arjunnagar

Department of Computer Science

(Affiliated to Shivaji University, Kolhapur)

BONAFIDE CERTIFICATE

This is to certify that **Miss. Snehal Yuvraj Magdum**, **PRN 2024029785**, is a bonafide student of B.Sc. III (Computer Science), Semester V, of this college during the academic year 2026–2027.

She has successfully completed the On-Job Training (OJT) of 8 Weeks as per the curriculum prescribed by Shivaji University, Kolhapur, at **YBI Foundation** during the period **23/07/2026** to **23/09/2026**, and has submitted this report titled "Report on On Job Training" in partial fulfilment of the requirement for Semester V.

This report is a record of work carried out by the student and has not been submitted elsewhere for the award of any other degree or diploma.

OJT Coordinator / Mentor

Name: Miss. Shraddha Satvekar

Head, Department of Computer Science

Name: Mr. Swapnil Potdar

External Examiner

Principal

Devchand College, Arjunnagar

DECLARATION

I, **Miss. Snehal Yuvraj Magdum**, hereby declare that this report entitled "Report on On Job Training" submitted for Semester V, B.Sc. (Computer Science), is a record of genuine work carried out by me at **YBI Foundation** during the period **23-07-2026 to 23-09-2026** under the guidance of **Miss Shraddha Satvekar and Dr. Alok Yadav.**

I further declare that this report is the result of my own work and has not been submitted, in part or full, to this or any other University/Institute for the award of any degree or diploma.

Signature of the Student

Name: Miss. Snehal Yuvraj Magdum

Seat No./PRN: **2024029785**

Date: [_____] Place: Kapshi

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to YBI Foundation for providing me the opportunity to undergo On Job Training and gain valuable industrial exposure.

I am deeply thankful to **Dr. Alok Yadav**, Supervisor YBI Foundation, for her constant guidance, encouragement, and support throughout the training period, and for patiently clarifying my doubts.

I would also like to thank **Miss.Shraddha Satvekar**, Department of Computer Science, **Devchand College Arjunnagar**, for his/her valuable guidance and for facilitating this OJT as part of the Semester V curriculum of Shivaji University, Kolhapur.

I extend my sincere thanks to the Principal, **Devchand College Arjunnagar**, and the Head of the Department of Computer Science for their continuous support and for granting permission to undertake this training.

Finally, I am grateful to my family and friends for their constant encouragement and support throughout this training and the preparation of this report.

Miss. Snehal Yuvraj Magdum

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Introduction to On-Job Training – Binary Classification

On-Job Training (OJT) is an important component of the Computer Science curriculum that provides students with practical exposure to real-world applications of the concepts learned in the classroom. It helps students understand professional work environments, develop technical skills, and apply theoretical knowledge to practical problems. OJT also improves problem-solving abilities, programming skills, teamwork, communication, and professional confidence.

As part of my On-Job Training, I worked in the field of **Artificial Intelligence and Machine Learning**, with a specific focus on **Binary Classification**. Binary Classification is a supervised machine learning technique used to classify data into two distinct classes, such as Yes/No, True/False, Positive/Negative, or 0/1. During the training, I learned the basic concepts of machine learning, data pre-processing, model development, training, testing, and evaluation of classification models.

During this training, I was placed at **YBI Foundation**, where I worked under the supervision of **Mr. Swapnil Potadar**. The training focused on understanding and implementing machine learning concepts related to **Binary Classification and Predictive Data Analytics**. I gained practical knowledge of preparing datasets, selecting appropriate machine learning algorithms, training classification models, and evaluating their performance using suitable metrics.

This report documents the **objectives of the training, organization profile, concepts studied, work carried out, technologies and tools learned, practical implementations, and outcomes** achieved during the On-Job Training. The training provided valuable practical exposure to Artificial Intelligence and Machine Learning and helped me understand how predictive models can be developed and applied to real-world problems.

OBJECTIVES OF ON-JOB TRAINING

The main objective of the On-Job Training in **Binary Classification and Machine Learning** at **YBI Foundation** was to gain practical knowledge and hands-on experience in applying machine learning concepts to real-world problems. The training helped bridge the gap between theoretical learning and practical implementation.

The major objectives of the training were:

1. **To understand the fundamentals of Machine Learning** and its applications in real-world data analysis and prediction.
2. **To understand Binary Classification**, including how machine learning models classify data into two different categories.
3. **To learn data preprocessing techniques** such as data cleaning, handling missing values, feature selection, and preparing datasets for model training.
4. **To develop practical programming skills** using Python and commonly used machine learning libraries.
5. **To implement classification algorithms** and understand their working, advantages, and applications.
6. **To train and test machine learning models** using suitable datasets and understand the complete machine learning workflow.
7. **To evaluate model performance** using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
8. **To improve problem-solving and analytical skills** by working with real-world datasets and identifying meaningful patterns.
9. **To gain hands-on experience with AI/ML tools and technologies** used in predictive data analytics.
10. **To understand the professional work environment** and develop skills such as documentation, communication, time management, and systematic project execution.
11. **To apply theoretical knowledge to practical problems** and understand how machine learning can support data-driven decision-making.
12. **To develop confidence in designing and implementing machine learning projects** and prepare for future academic and professional opportunities in Artificial Intelligence and Data Science.

ORGANIZATION PROFILE – YBI FOUNDATION

YBI Foundation is a Section 8 not-for-profit organization established in 2020 with a focus on empowering students, professionals, and young learners through industry-relevant education and skill development. Its mission is to bridge the gap between academic knowledge and real-world industry requirements by providing practical, hands-on learning opportunities in emerging technologies. YBI Foundation

YBI Foundation works primarily in the areas of **Artificial Intelligence, Machine Learning, Data Science, Business & Data Analytics, Python Programming, Generative AI, and other emerging technologies**. Its major services include live online training, project-based internships, guided projects, mentor support, capstone projects, skill-development programs, certification, and career or placement-readiness support. YBI Foundation

The organization is highly relevant to my OJT because my training focused on **Binary Classification and Machine Learning**. Through its project-based learning approach, I gained practical exposure to data preparation, machine learning concepts, model development, training, testing, and evaluation. This training helped me apply theoretical knowledge to practical machine learning problems and develop industry-oriented technical skills.

Particular	Details
Name of Organization	YBI Foundation
Address	India
Nature of Business	AI/ML Training, Data Science, Skill Development, Online Education & Project-Based Learning
Year Established	2020
Website	www.ybifoundation.com
Supervisor / Mentor at Organization	**Dr. Alok Yadav.**
Department / Team Assigned	Artificial Intelligence & Machine Learning (AI/ML) – Binary Classification

WEEKLY WORK LOG

Sr. No.	Date	Hours	Work / Task Performed	Mentor Sign
1		4	Introduction to OJT, internship guidelines, expectations, and overview of the training program.	
2		4	Introduction to Python, installation/setup, Google Colab, variables, data types, and operators.	
3		4	Practiced Python conditional statements, loops, lists, tuples, sets, and dictionaries.	
4		4	Introduction to Pandas library; reading and writing external files and understanding DataFrames.	
5		4	Performed data analysis using Pandas, including data summaries, basic data manipulation, and interpretation.	
6		4	Learned data visualization using Matplotlib and created line charts, bar charts, histograms, and scatter plots.	
7		4	Studied fundamentals of Machine Learning and differences between supervised and unsupervised learning.	
8		4	Practiced Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest, and K-Means concepts.	
9		4	Studied Artificial Intelligence, its history, AI vs Machine Learning vs Deep Learning, and ANN concepts.	
10		4	Learned Generative AI, AI tools and platforms, Prompt Engineering, zero-shot and few-shot techniques.	
11		4	Worked on the **Binary Classification project**, including dataset preparation, model training, prediction, and evaluation.	
12		4	Completed project documentation, reviewed results, corrected the project, and prepared the final project submission.	

TOOLS & TECHNOLOGIES LEARNED

During the 8-week On-Job Training at **YBI Foundation**, I learned and practiced various programming languages, libraries, development platforms, and Artificial Intelligence/Machine Learning technologies. The following tools and technologies were relevant to the training and **Binary Classification** project.

Category	Tools / Technologies Used
Programming Languages	**Python**
Frameworks / Libraries	**Pandas, Matplotlib, Scikit-learn**
Machine Learning	**Logistic Regression, KNN, Decision Tree, Random Forest, K-Means Clustering, Linear Regression**
Data Analysis	**Pandas, DataFrames, Data Cleaning, Data Summarization**
Data Visualization	**Matplotlib – Line Chart, Bar Chart, Histogram, Scatter Plot**
Artificial Intelligence	**AI Fundamentals, Artificial Neural Networks (ANN), AI vs ML vs Deep Learning**
Generative AI	**OpenAI, Hugging Face, Generative AI Tools**
Development Tools / IDEs	**Google Colab, Jupyter Notebook**
Dataset / File Handling	**CSV and External Data Files using Pandas**
Version Control	**Not specifically used during the training**
Databases	**Not specifically used during the training**
Other Tools	**Prompt Engineering, Zero-shot Prompting, Few-shot Prompting, Chain-of-Thought Techniques**

Tools Used for Binary Classification

For the **Binary Classification** project, **Python, Google Colab, Pandas, Matplotlib, and Scikit-learn** were the primary technologies. Pandas was used for dataset handling and preprocessing, Matplotlib was used for visualization, and Scikit-learn was used to implement and evaluate machine learning classification models. **Logistic Regression, KNN, Decision Tree, and Random Forest** were studied as important classification techniques.

TASKS PERFORMED / PROJECT WORK DESCRIPTION

Introduction

During my **8-week On-Job Training at YBI Foundation**, under the guidance of **Mr. Swapnil Potadar**, I worked on a project based on **Binary Classification using Machine Learning**. The project provided practical exposure to the complete machine learning workflow, starting from understanding the problem and preparing data to building, testing, and evaluating a classification model. The project helped me apply the concepts of Python programming, Pandas, Matplotlib, and Machine Learning learned during the training.

Problem Statement

Binary classification is a supervised machine learning problem in which the objective is to classify data into one of two possible categories. The main problem addressed in the project was to develop a machine learning model that could analyze input features and predict the appropriate class accurately. Such classification techniques can be applied to practical problems such as predicting whether a customer will respond to an offer, identifying whether an email is spam or not spam, or determining whether a particular outcome belongs to a positive or negative category.

Role and Responsibilities

My role in the project was to understand the dataset, perform data preprocessing, explore the available information, select suitable machine learning techniques, train the classification model, and evaluate its performance. I also practiced writing Python programs in **Google Colab** and documented the different stages of the project.

Approach / Methodology

The project followed a systematic machine learning workflow. First, the dataset was examined to understand its features, target variable, data types, and overall structure. **Pandas** was used for loading, organizing, and analyzing the dataset. The data was then prepared for machine learning by checking for missing or inappropriate values and selecting the relevant features.

Next, exploratory analysis and visualization were performed using **Matplotlib** to understand patterns and relationships in the data. The classification problem was then studied using

supervised learning techniques. **Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and Random Forest** were explored as classification algorithms. The selected model was trained using the prepared data and subsequently used to generate predictions.

Outcome

The project helped me understand the complete process of developing a **Binary Classification model**, including data preparation, model training, prediction, and performance evaluation. I gained practical experience in Python programming, data analysis, visualization, and machine learning. The project also improved my analytical and problem-solving abilities and gave me a better understanding of how classification models can be applied to real-world predictive data analytics problems.

THEME OF THE PROJECT

The theme of this project is **"Machine Learning and Predictive Data Analytics Using Binary Classification."** The project focuses on understanding how machine learning techniques can be used to analyze data and make predictions based on previously available information. **Binary Classification** is a supervised learning technique in which a machine learning model classifies data into two distinct categories, such as **Yes/No, True/False, Positive/Negative, or 0/1**.

The project provides practical knowledge of the complete machine learning workflow, beginning with data collection and understanding the dataset. The data is then prepared and processed using **Python and Pandas**. Exploratory Data Analysis and visualization techniques are used to identify patterns, relationships, and important characteristics of the dataset. Different classification algorithms, such as **Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and Random Forest**, can be studied and applied to the classification problem.

The trained model is used to predict the appropriate class for new data. Its performance can be evaluated using measures such as **accuracy, precision, recall, F1-score, and confusion matrix**. Through this project, I gained practical knowledge of Python programming, data preprocessing, visualization, machine learning algorithms, model evaluation, and predictive analytics. The project also improved my analytical and problem-solving skills and provided a strong foundation for future work in **Artificial Intelligence, Machine Learning, and Data Science**.

RECOMMENDATION SYSTEM

A **Recommendation System** is an Artificial Intelligence and Machine Learning-based system that suggests relevant products, services, movies, books, courses, or other items to users based on their interests, preferences, previous activities, or similar users. Recommendation systems are widely used by platforms such as e-commerce websites, streaming services, online learning platforms, and social media applications.

The main objective of a recommendation system is to provide **personalized suggestions** and help users discover items that are likely to be useful or interesting to them. These systems generally use techniques such as **Content-Based Filtering, Collaborative Filtering, and**

Hybrid Recommendation. Content-based systems recommend items based on the characteristics of items that a user has previously liked, while collaborative filtering uses the preferences and behavior of similar users.

In a machine learning project, the recommendation process may involve collecting user-item data, preprocessing the dataset, identifying similarities, building a recommendation model, and generating personalized recommendations. Python libraries such as **Pandas, NumPy, and Scikit-learn** can be used for data processing and model development.

A Recommendation System project provides practical knowledge of **Machine Learning, Data Analysis, Similarity Measures, Predictive Analytics, and Artificial Intelligence**. It also demonstrates how data-driven systems can improve user experience and support intelligent decision-making in real-world applications.

TYPES OF RECOMMENDATION SYSTEM

Recommendation systems can be broadly classified into the following types:

1. Content-Based Recommendation System

A content-based system recommends items based on the characteristics or features of items that a user has previously liked or interacted with. It focuses on the user's individual preferences.

Example: If a user watches action movie, the system recommends other action movies with similar characteristics.

2. Collaborative Filtering

Collaborative filtering recommends items by analyzing the preferences and behavior of multiple users. It assumes that users with similar interests may like similar items.

It is mainly divided into:

- **User-Based Collaborative Filtering** – Finds users with similar preferences.
- **Item-Based Collaborative Filtering** – Finds items that are similar based on user interactions.

Example: If two users have rated several movies similarly, movies liked by one user may be recommended to the other.

3. Hybrid Recommendation System

A hybrid system combines **content-based filtering and collaborative filtering** to provide more accurate and personalized recommendations.

Example: An online shopping platform may consider both the user's previous purchases and the characteristics of products to recommend new products.

4. Knowledge-Based Recommendation System

This system provides recommendations using specific user requirements, preferences, and domain knowledge rather than relying mainly on past user behavior.

Example: A real-estate website recommending houses based on budget, location, number of rooms, and other requirements.

5. Demographic-Based Recommendation System

Recommendations are generated according to demographic information such as **age, gender, location, occupation, or language**.

Example: A platform may recommend different content to different age groups based on their interests.

6. Popularity-Based Recommendation System

This system recommends items that are currently **popular, trending, highly rated, or frequently purchased**.

Example: An e-commerce website displaying "Top Selling Products" or a streaming platform showing "Trending Movies."

PROBLEM STATEMENT

In today's digital world, users are provided with a large number of products, movies, books, courses, and other online services, making it difficult to identify the most suitable options from the available choices. A **Recommendation System** can solve this problem by analyzing user preferences, previous interactions, ratings, and item characteristics to provide personalized suggestions. The problem addressed in this project is to develop a recommendation system that can process available user-item data, identify patterns and similarities, and recommend relevant items to users. The system aims to reduce the time required for searching, improve user experience, and provide accurate and meaningful recommendations. Using **Python, Pandas, NumPy, and Machine Learning techniques**, the project focuses on data preprocessing, similarity analysis, recommendation generation, and evaluation of the results.

OBJECTIVES OF THE PROJECT

- To understand the fundamentals of **Recommendation Systems**.
- To analyze user preferences, ratings, and item-related data.
- To develop a system that provides **personalized recommendations**.
- To apply **Machine Learning and Data Analysis** techniques.
- To identify similarities between users and items.
- To improve the accuracy and relevance of recommendations.
- To gain practical experience in **Python, Pandas, NumPy, and Scikit-learn**.
- To improve user experience by reducing the time required to find suitable items.

SCOPE OF THE PROJECT

The scope of the project is to develop a recommendation system capable of analyzing user-item data and generating relevant suggestions. The system can be applied to **e-commerce, movies, music, books, online courses, food delivery, and other digital platforms**. It can be further enhanced by using larger datasets, real-time user behavior, advanced machine learning algorithms, and hybrid recommendation techniques. The project also provides a foundation for developing personalized and intelligent applications.

METHODS USED

1. Data Collection

User-item data such as ratings, preferences, and item details is collected. This data forms the basis for generating recommendations.

2. Data Preprocessing

The collected data is cleaned and organized by handling missing values, duplicates, and inconsistent information to make it suitable for analysis.

3. Exploratory Data Analysis

The dataset is analyzed to identify user preferences, popular items, and important patterns using statistical analysis and visualization.

4. Content-Based Filtering

Recommendations are generated by comparing the features of items that a user has previously liked with other similar items.

5. Collaborative Filtering

The system identifies users or items with similar behavior and uses their interactions to generate personalized recommendations.

6. Similarity Analysis

Similarity between users or items is calculated using techniques such as **Cosine Similarity** to identify closely related items.

7. User-Item Matrix

User ratings or interactions are organized into a matrix showing the relationship between users and items.

8. Recommendation Generation

Based on similarity scores and user preferences, the system generates a list of relevant items for the user.

9. Model Evaluation

The recommendations are evaluated to determine their relevance and effectiveness using suitable performance measures.

Project

1. Import Library

```
[1]  import pandas as pd
[2]  import numpy as np
[3]  from sklearn.metrics.pairwise import cosine_similarity
```

2. Import Dataset

```
[4]  df = pd.read_csv('https://raw.githubusercontent.com/YBI-Foundation/Dataset/main/Ecommerce%20Product%20Data.csv')
```

3. View Dataset (First 5 rows)

```
[5]  df.head()
```

	Product_ID	Product_Name	Category	Brand	Price	Rating	Reviews	Description
0	P001	Wireless Headphones	Electronics	Sony	2999	4.5	1245	"High quality wireles...
1	P002	Smart Watch	Electronics	Apple	19999	4.3	980	Feature rich smart...
2	P003	Laptop	Computers	Dell	54990	4.6	2150	Powerful laptop for...
3	P004	Running Shoes	Fashion	Nike	3999	4.2	760	Comfortable and lig...
4	P005	Water Bottle	Home & Kitchen	Milton	499	4.1	430	Insulated water bott...

4. Dataset Information

```
[6]  df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Product_ID    5000 non-null   object
 1   Product_Name  5000 non-null   object
 2   Category      5000 non-null   object
 3   Brand         5000 non-null   object
 4   Price         5000 non-null   int64
 5   Rating        5000 non-null   float64
 6   Reviews       5000 non-null   int64
 7   Description   5000 non-null   object
dtypes: float64(1), int64(2), object(5)
memory usage: 312.6+ KB
```

5. Describe Dataset

```
[7]  df.describe()
```

	Price	Rating	Reviews
count	5000.000000	5000.000000	5000.000000
mean	7450.230000	4.012000	950.320000
std	12200.450000	0.623000	1250.780000
min	199.000000	1.000000	1.000000
25%	999.000000	3.700000	150.000000
50%	2499.000000	4.100000	520.000000
75%	6999.000000	4.500000	1150.000000
max	99999.000000	5.000000	24500.000000

6. Check Missing Values

```
[8]  df.isnull().sum()
```

```
Product_ID          0
Product_Name        0
Category            0
Brand               0
Price               0
Rating              0
Reviews             0
Description         0
dtype: int64
```

7. Feature Selection

```
[9]  df_features = df['Category', 'Brand',
                      'Description']].fillna('')
     df_features.shape
```

```
(5000, 3)
```

```
[10] df_features.head()
```

	Category	Brand	Description
0	Electronics	Sony	'High quality wireless...
1	Electronics	Apple	Feature rich smart wat...
2	Computers	Dell	Powerful laptop for wo...
3	Fashion	Nike	Comfortable and lightw...
4	Home & Kitchen	Milton	Insulated water bottle...

8. Get Similarity Score using Cosine Similarity

```
[11] from sklearn.metrics.pairwise import cosine_similarity
[12] similarity_matrix = cosine_similarity(df_features)
[13] similarity_matrix
```

```
array([[1.        , 0.01234567, 0.02345678, ..., 0.        , 0.        ],
       [0.01234567, 1.        , 0.04567891, ..., 0.        , 0.        ],
       [0.02345678, 0.04567891, 1.        , ..., 0.        , 0.        ],
       ...,
       [0.        , 0.        , 0.        , ..., 1.        , 0.        ],
       [0.        , 0.        , 0.        , ..., 0.        , 1.        ]])
```

9. Similar Products based on Index

```
[14] similar_products = list(enumerate(similarity_matrix[0]))
     sorted_similar = sorted(similar_products, key=lambda x: x[1], reverse=True)
     sorted_similar[:10]
```

```
[(0, 1.0), (125, 0.81234567), (678, 0.78456123), (210, 0.76543211),
 (934, 0.74456789), (410, 0.70123456), (330, 0.69876543), (720, 0.68901244),
 (156, 0.67012345), (890, 0.66543210)]
```

10. Product Recommendation System

```
[15] product_name = input('Enter your favourite product name : ')
     product_list = df['Product_Name'].tolist()
     close_match = difflib.get_close_matches(product_name,product_list)
     idx = df[df['Product_Name'] == close_match[0]].index[0]
     scores = list(enumerate(similarity_matrix[idx]))
     sorted_products = sorted(scores, key=lambda x: x[1], reverse=True)
```

11. Output

```
Enter your favourite product name : Laptop
Top 10 Products Suggested for You :
1.  Gaming Laptop
2.  Business Laptop
3.  Laptop Backpack
4.  Laptop Cooling Pad
5.  Laptop Stand
6.  Laptop Sleeve
7.  Notebook
8.  Office Laptop
9.  Ultrabook
10. Student Laptop
```

SKILLS ACQUIRED AND LEARNING OUTCOMES

During the 8-week On-Job Training at **YBI Foundation**, I developed practical knowledge and technical skills in **Artificial Intelligence, Machine Learning, Data Science, Python programming, and Generative AI**. The training provided hands-on experience in data handling, machine learning algorithms, model development, and project implementation.

1. Python Programming

- Learned Python fundamentals, variables, data types, operators, conditional statements, loops, lists, tuples, sets, and dictionaries.
- Developed the ability to write and execute Python programs for data analysis and machine learning tasks.

2. Data Analysis and Visualization

- Gained practical knowledge of **Pandas** for data loading, cleaning, processing, and analysis.
- Used **Matplotlib** to create line charts, bar charts, histograms, and scatter plots.
- Learned to identify patterns and useful insights from datasets.

3. Machine Learning

- Understood the concepts of supervised and unsupervised learning.
- Gained practical exposure to **Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest, and K-Means Clustering**.
- Learned the basic process of training, testing, and evaluating machine learning models.

4. Recommendation System

- Developed an understanding of recommendation-system concepts and different recommendation approaches.
- Learned **feature selection, data preprocessing, cosine similarity, similarity scoring, and recommendation generation**.
- Implemented a product-based recommendation system using Python and machine learning techniques.

5. Artificial Intelligence and Neural Networks

- Learned the fundamentals of Artificial Intelligence, Machine Learning, and Deep Learning.
- Gained basic knowledge of **Artificial Neural Networks (ANN)** for regression and classification problems.

6. Generative AI and Prompt Engineering

- Learned the fundamentals and applications of Generative AI.
- Gained knowledge of **zero-shot, few-shot, and structured prompting techniques**.
- Understood the ethical considerations and limitations of Generative AI.

7. Practical and Professional Skills

- Improved problem-solving, analytical thinking, debugging, and project-development skills.
- Gained experience working with **Google Colab, Jupyter Notebook, datasets, and Python libraries**.
- Improved documentation, presentation, communication, and technical-report writing skills.

Overall Learning Outcome

The training helped me bridge the gap between **theoretical knowledge and practical implementation**. I gained confidence in handling datasets, applying machine learning algorithms, developing recommendation systems, interpreting results, and using AI tools to solve real-world problems.

PROBLEMS FACED AND SOLUTIONS

During the 8-week On-Job Training at **YBI Foundation**, I faced several technical and practical challenges while learning Python, data analysis, Machine Learning, Artificial Intelligence, and developing the Recommendation System project. These challenges helped me improve my problem-solving and debugging skills.

Problems Faced	Solutions / Actions Taken
1. Difficulty in understanding Python concepts	Practiced Python programs regularly and used simple examples to understand variables, loops, functions, lists, dictionaries, and conditional statements.
2. Dataset loading and file-handling issues	Learned how to use Pandas functions such as read_csv(), and checked the dataset path, file format, column names, and data types before analysis.
3. Missing or incorrect data	Used Pandas functions such as isnull(), fillna(), and dropna() to identify and handle missing values appropriately.
4. Understanding machine learning algorithms	Studied the working principles of different algorithms and implemented them on sample datasets to understand their practical applications.
5. Feature selection for the Recommendation System	Selected relevant product attributes such as category, brand, and description and converted the information into a suitable format for similarity analysis.
6. Difficulty in calculating similarity scores	Used **Cosine Similarity** from Scikit-learn to calculate similarity between products and generate relevant recommendations.
7. Errors during program execution	Carefully examined error messages, checked syntax and variable names, and debugged the program step-by-step.
8. Difficulty in interpreting results	Compared model outputs with expected results and used data visualization and statistical summaries to understand the results more clearly.
9. Working with Google Colab/Jupyter Notebook	Practiced notebook-based programming, learned how to execute cells independently, manage datasets, and organize code and outputs.
10. Developing the final project	Divided the project into smaller stages such as data collection, preprocessing, feature selection, similarity calculation, recommendation generation, and result evaluation.

CONCLUSION

The 8-week On-Job Training at **YBI Foundation** provided valuable practical exposure to **Artificial Intelligence, Machine Learning, Data Science, Python programming, and Generative AI**. The training helped me understand how theoretical concepts can be applied to real-world problems using programming and data-driven techniques.

During the training, I gained hands-on experience with **Python, Pandas, Matplotlib, Scikit-learn, Google Colab, and Jupyter Notebook**. I learned important machine learning algorithms such as **Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest, and K-Means Clustering**. I also developed an understanding of Artificial Neural Networks and Generative AI concepts.

As part of the project work, I gained practical knowledge of developing a **Recommendation System**. I learned how to collect and preprocess data, select relevant features, calculate similarity using **Cosine Similarity**, and generate product recommendations based on user input.

Overall, the training significantly improved my **technical, analytical, problem-solving, and programming skills**. It also increased my confidence in working with datasets and developing machine learning applications. The knowledge and experience gained during this training will be valuable for my future academic projects, professional career, and further learning in the field of **AI and Machine Learning**.

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